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ECA0502 EMF

EXP 2

Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

PROGRAM:

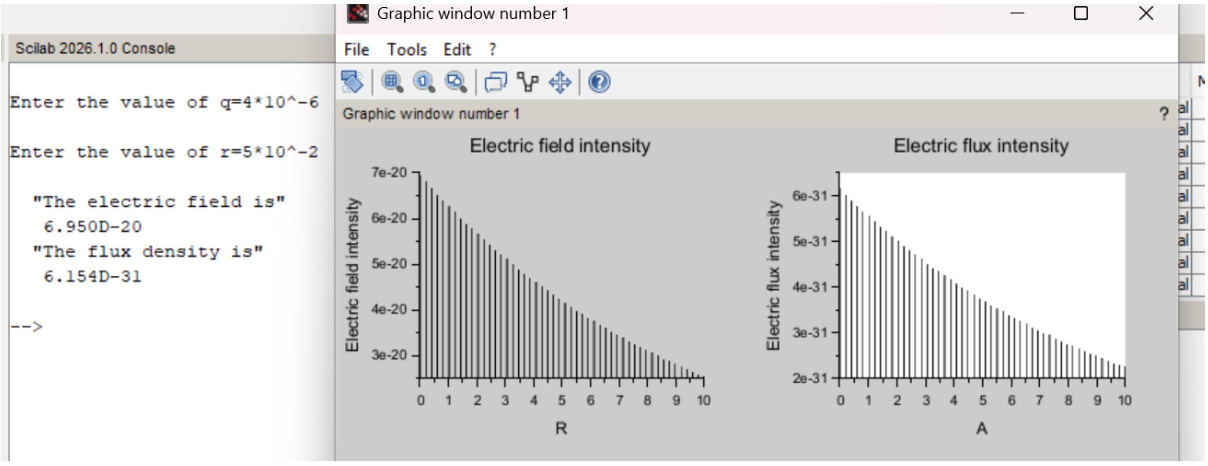
```
clc;
clear;

q=input('Enter the value of q=');
r=input('Enter the value of r=');

efcilon=8.854*10^-12;
pi=3.14
E=q/4*pi*efcilon*r^2;
disp("The electric field is", [E]);
D=E*efcilon;
disp("The flux density is", [D]);
x= linspace(0, 10,50);
y=E*exp(-0.1*x);
z=D* exp(-0.1*x);

figure();
subplot(2, 2,1);
plot2d3(x,y);
xlabel("R");
ylabel("Electric field intensity");
title('Electric field intensity');
subplot(2, 2, 2);
plot2d3(x,z);
xlabel("A");
ylabel("Electric flux intensity");
title('Electric flux intensity');
```

OUTPUT:



OBSERVATION:

Exp: 3

Force attraction between two electric charges and two parallel conductors

value of:

charge 1 = 2×10^{-6}

charge 2 = 3×10^{-6}

radius = 3 mm

current 1 = 3

current 2 = 4

length = 5 cm

d = 2 mm

$$F = \frac{q_1 q_2}{4} \pi \varepsilon_0 r^2 = \frac{6 \times 10^{-12}}{4} \times 3.14 \times 8.854 \times 10^{-12} \times 9 \times 10^{-6}$$

$$\boxed{F = 3.7532 \times 10^{-28}\ N}$$

o/p ✓

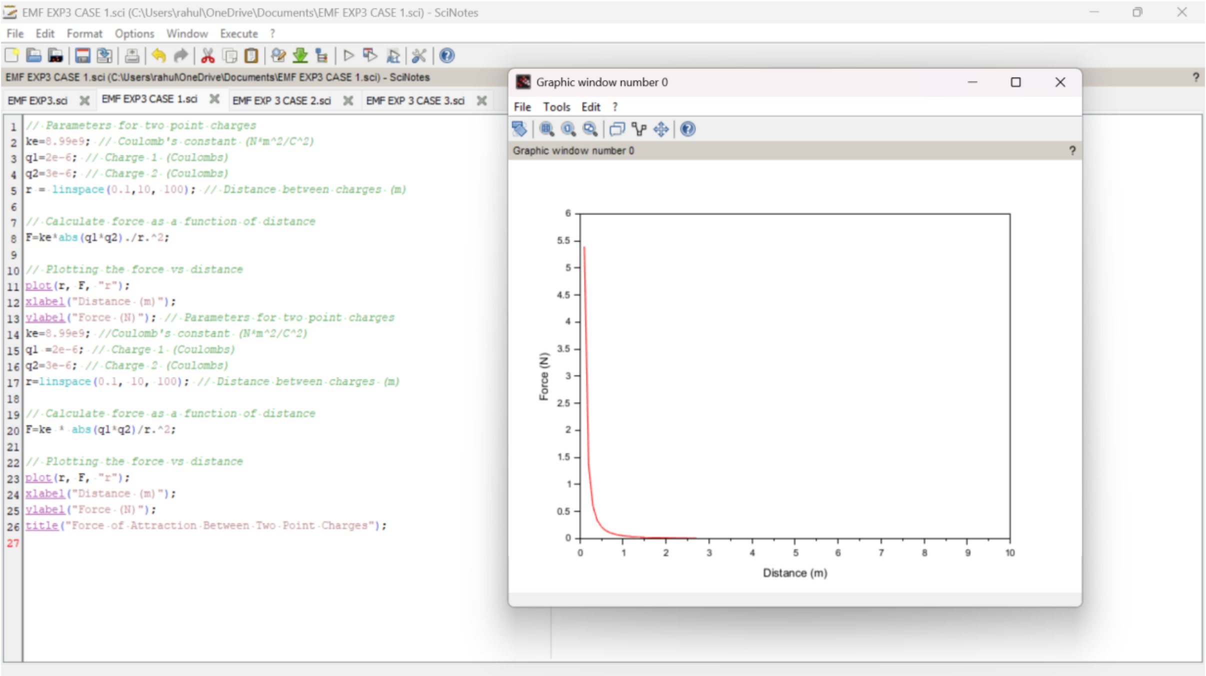
$$f = \frac{\mu I_1 I_2 l}{2} \times \pi \times d = \frac{7.536 \times 10^{-7}}{2} \times 3.14 \times 2 \times 10^{-3}$$

$f = \cancel{4}3$ ~~f = 4.73×10^-8~~

$$\boxed{f = 2.366 \times 10^{-9}}$$

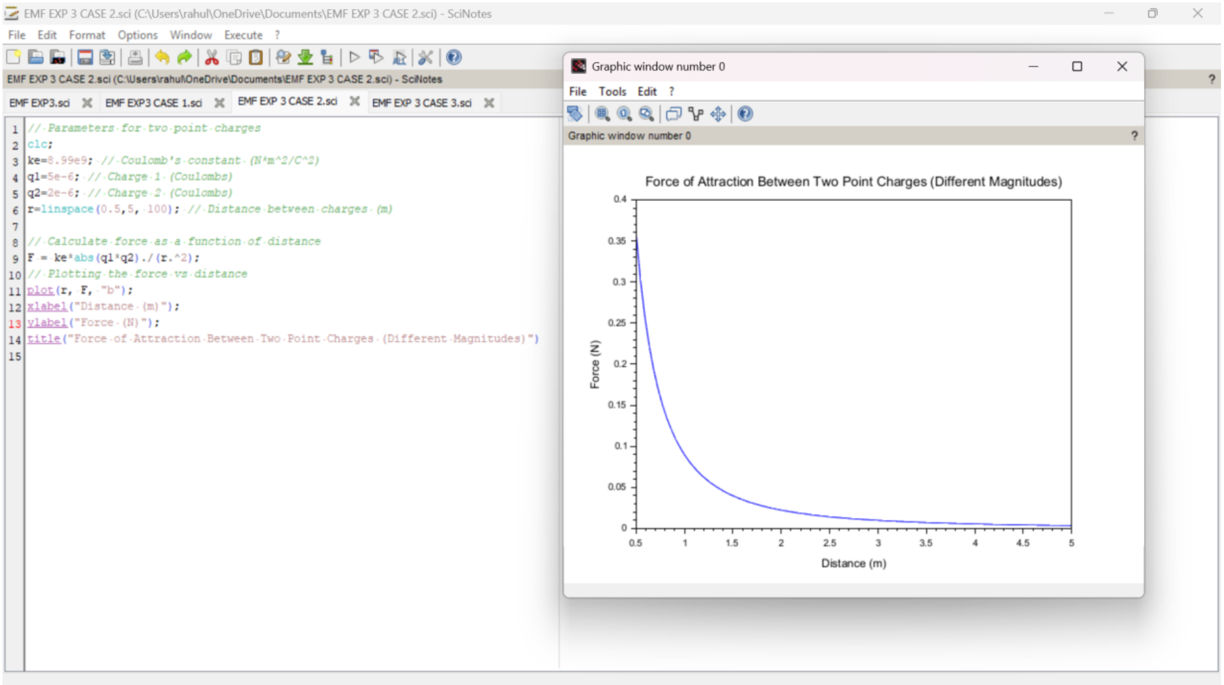
CASE 1:

PROGRAM AND OUTPUT:



CASE 2:

PROGRAM AND OUTPUT:



CASE 3:

PROGRAM AND OUTPUT